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PERNAMBUCO WOOD (*CAESALPINIA ECHINATA*) USED IN THE MANUFACTURE OF BOWS FOR STRING INSTRUMENTS

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SUMMARY

The anatomy and the physical, mechanical, acoustical and chemical properties of pernambuco wood (*Caesalpinia echinata* Lam.) were investigated to determine factors that could explain the different quality of sticks used in bow manufacture. Eighteen sticks were classified into four classes (A to D with A being the best) according to their potential quality for bow manufacture. Selection of samples was based on the experience of a bow maker and on some nondestructive tests. The A-class sticks had a lower frequency of vessels and rays and a higher percentage of fibers when compared to the other classes. They also had higher values of density, speed of sound propagation, modulus of elasticity and modulus of rupture. Klason lignin content was higher in the A-class sticks but the quantity of hydrosoluble and ethanol/benzene soluble extractives was lower. The values of density and speed of sound propagation obtained by nondestructive and destructive methods were similar showing the applicability of the former in the prior selection process of the sticks.

Key words: Violin bow, pernambuco wood, anatomy, *Caesalpinia echinata*.

INTRODUCTION

The process of manufacturing musical instruments can be considered a fusion of art and technology. The luthier is an artist who uses specific tools and techniques to give each instrument unique aesthetic and functional characteristics (Oliveira 1984). This also applies to the bow maker or archetier.

According to Retford (1964), the precise origin of the bows is unknown; the oldest references are from Arabic and Byzantine culture and date from the 10th century. Before 1800, musicians considered bows as mere accessories, but with the arrival of more sophisticated music, bow makers developed stronger and more responsive bows, and players realized that bows were just as important as the instruments in producing a better sound (Mnatzaganain 2002).

The making of bows began to be considered a specialized art in the 18th century with "Old Tourte" and his workshop in Paris. Before the middle of the 18th century, bow makers used various tropical hardwoods including pernambuco, established as the ideal material for the manufacturing of bows by Tourte's sons (François and Xavier). François Tourte (1747–1835), working in partnership with the violinist G.B. Viotti, gave the bow its modern shape and established its ideal length (between 73.3 and 75.0 cm) (Retford 1964).

The concept of quality of a bow is relative, but usually the good bow is the one that is appreciated by the musician and which harmonizes the instrument / musician / bow trio. The bow makers consider that the length, the weight (for the violin, between 60 and 64 g) and the curve, which is considered its "soul", must be respected when evaluating the bow quality (Retford 1964).

Based on the experience of bow makers and musicians, pernambuco wood (*Caesalpinia echinata* Lam. – Fabaceae) has been considered the best wood for bows for more than 200 years because it combines ideal characteristics of resonance, density, durability and beauty, among others (Bueno 2002). Nevertheless, practice shows that pernambuco wood sticks processed in the same way by the same bow maker can result in bows with very distinct qualities.

This variation is due to a series of factors that probably involve the anatomy of the wood and its physical, mechanical, acoustic and chemical properties. With this in mind, some bow makers use their experience to make a prior selection of the sticks before they begin the process of manufacturing the bows. Nevertheless, some objective parameters such as wood density and the speed of sound propagation reduce the guesswork involved in evaluating the potential tonal properties of a stick. Using these parameters it is possible to calculate the modulus of elasticity (MOE) which is the measure of resistance of the wood to deflection. Many modern bow makers consider this when buying pernambuco blanks for bows (Angyalossy *et al.* 2005, 2006).

There are no scientific studies that relate the structure and the physical, mechanical, acoustic and chemical properties of pernambuco wood to bow quality, so the objective of this study was to explore the possibility of establishing such correlations and to define which are the anatomical characteristics and the physical, mechanical, acoustical and chemical properties that give the potential quality to the pernambuco wood to be used in bow making.

MATERIALS AND METHODS

Sample selection

About 150 sticks (740 × 45 × 25 mm) were obtained from 10 blanks of pernambuco wood (995 × 90 × 50 mm). From these, 18 sticks were selected and classified into four classes according to their potential quality for the manufacture of bows, and were designated as A class (high), B class (good), C class (intermediate) and D class (low). For the selection of the sticks, the wood density and the speed of sound propagation through the wood were taken in consideration (Table 1). The density was determined from the relation between the stick mass and the volume of water displaced when it was immersed in a graduated cylinder containing water. The speed of sound propagation was determined using the Lucchi Meter. This is an ultrasonic tester designed to measure the acoustic characteristics of a piece of wood. Specifically it measures the time needed for sound waves to travel through the material. This information can then be used to determine the elasticity of the wood, as well as its sound quality. The elasticity of a material can be obtained by the equation: Elasticity = $(\text{velocity})^2 \times$ density (Lucchi 1986).

Table 1. Classification of wood sticks following the density at 12 percent r.h. and the sound speed. Classes represent the potential quality of the bow: A = high; B = good; C = intermediate; D = low.

Stick number	Density (kg m^{-3})	Speed of sound (m s^{-1}) Lucchi Tester	Classes
1	1040	5440	A
2	1030	5480	A
3	1020	5440	A
4*	1000	5320	A
5*	1000	5390	A
6*	1010	5410	A
1	990	5410	B
2	990	5280	B
3	990	5210	B
1	1010	5110	C
2	990	5170	C
3	980	5160	C
1	1080	5000	D
2	1010	4910	D
3	850	5060	D
4*	980	4630	D
5	#	3900	D
6*	830	4990	D

* Values also determined according NBR 7190/97 (see Table 4).
\# no data available.

Microscopic features

Two blocks (1.5 cm^3) were cut from the ends of the 18 samples. The blocks were softened in boiling water and glycerin (4 : 1). A sliding microtome was used to cut transverse and longitudinal sections 15–20 μm thick. The sections were cleared with sodium hypochlorite (60 %), washed in water, stained with astrablue and safranin (aqueous, 1%, 9 : 1) (Johansen 1940) and mounted in Permount®. Macerations were prepared using the modified Franklin's method (Berlyn & Miksche 1976), stained with astrablue and safranin and mounted in a 50 % solution of glycerin and water. For the A and D classes, the proportions of axial parenchyma, fibers, rays and vessels in the transverse sections were measured using a 25-point grid on 60 successive areas. Terminology followed the IAWA list (IAWA Committee 1989).

Physical, mechanical and acoustic properties

Destructive tests were performed on six sticks, three of the A class and three of the D class, according to the Brazilian standards of material characterization (Brazilian norm NBR 7190/97 – Wood Structure Project of ABNT - Associação Brasiliera de Normas Técnicas (Brazilian Association of Technical Norms). The density at 12 % humidity, elastic modulus, the modulus of rupture and speed of sound propagation were measured.

Density was determined using 36 samples (27 × 17 × 15 mm) taken from both ends and from the central region of the sticks marked with (*) in Table 1. Their mass was measured using an analytical balance, and they were immersed in mercury to determine the displaced volume. The density value for each stick corresponds to the average of the three regions of the analyzed stick.

Sticks (745 × 17 × 15 mm) were tested in a Universal Test Machine (Instron) for the determination of their mechanical properties. Each stick was placed in the machine and received increasing loads in the central region; the resulting deformities were recorded until the proportionality limit was reached. The process proceeded until the sticks were broken and the resistance of each stick was established. Upon knowing the values of the proportionality limit, the rupture force, the dimensions of the sticks and the support interspaces, it was possible to calculate the modulus of rupture and the modulus of elasticity (Clough & Penzien 1982).

The speed of sound propagation was established based on the modulus of elasticity and the wood density (Bucur 1995; Yojo 2004).

Chemical properties

Klason lignin and extractive content of the A and D classes were determined by analysis of the stick fragments evaluated in the destructive tests.

For determination of the Klason lignin content we followed the methodology described by Hatfield *et al*. (1994). Samples with 2 g powder were passed several organic solvents, acid and hydroalcoholic solutions. In addition, samples were washed and centrifugated, and the pellet was quantified.

Analysis of the extractives (3 g powder samples) was carried out in water and in ethanol-benzene solution using the method of Matsunaga *et al*. 1996.

RESULTS

Wood anatomy

The sticks with higher potential (A and B classes) had a lower vessel frequency when compared with the sticks of C and D classes (Fig. 1–8). D-class sticks had longer vessel elements confirmed by statistical analysis (Table 2). The tangential diameter of the vessels did not vary significantly among the sticks of the different classes. A- and B-class sticks had fibers with larger than average lumens, although cell wall thickness did not vary among the samples from the different classes. The length and the width of the fibers did not show any correlation with the classes. The same was observed with regard to the height and width of the rays. The frequency of rays was lower in the A- and B-class sticks (Table 2). Storied structure was present in some sticks; in others it was less visible or even absent. It was not possible to correlate this variation to the stick quality (Fig. 9 & 10).

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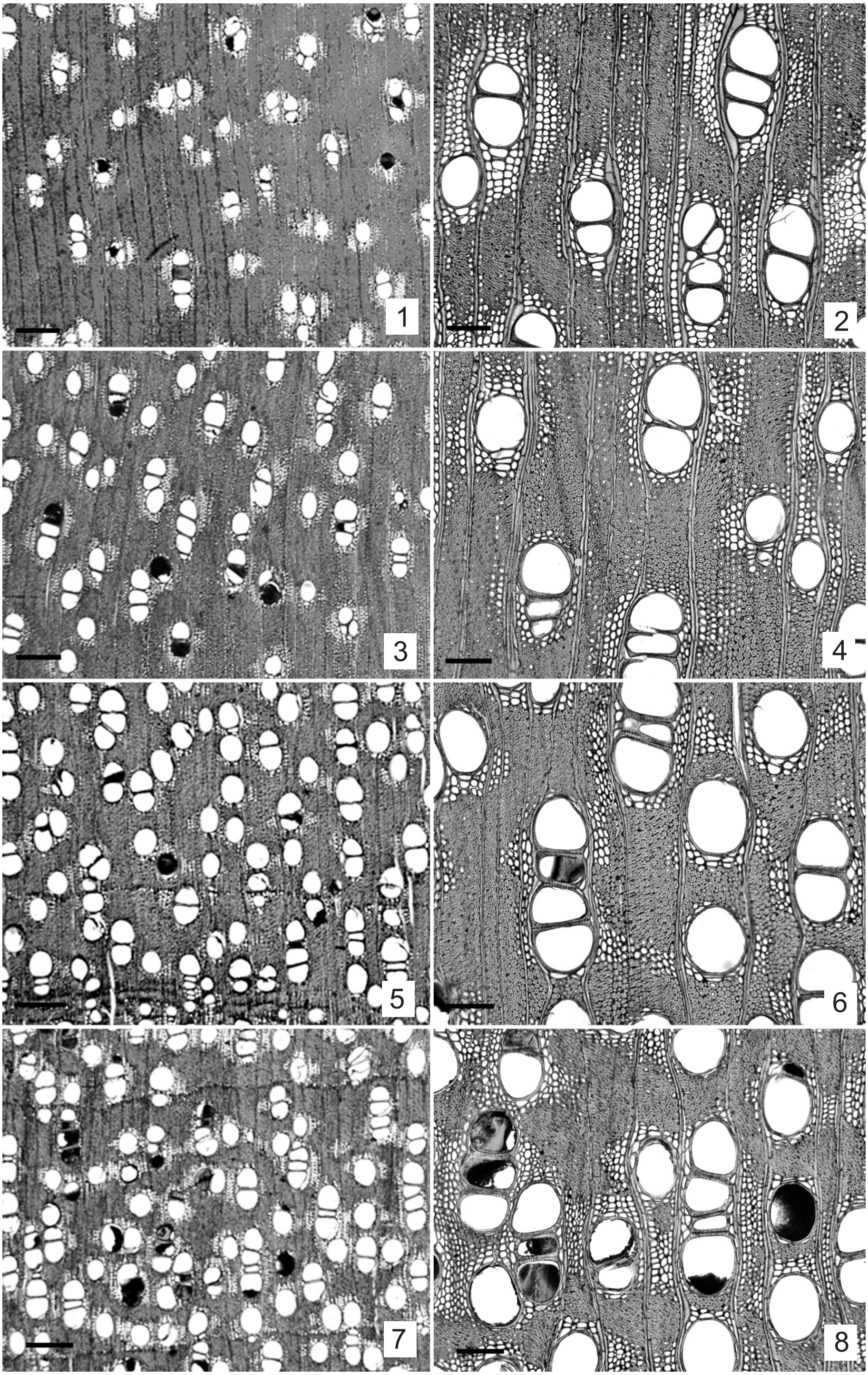
Figures 1–8. Transverse sections of *Caesalpinia echinata* wood sticks showing variation of vessel frequency. – 1 & 2. A-class sticks. – 3 & 4. B-class sticks. – 5 & 6: C-class sticks. – 7 & 8. D-class sticks. — Scale bars: 500 μm in Fig. 1, 3, 5, 7; 100 μm in Fig. 2, 4, 6, 8.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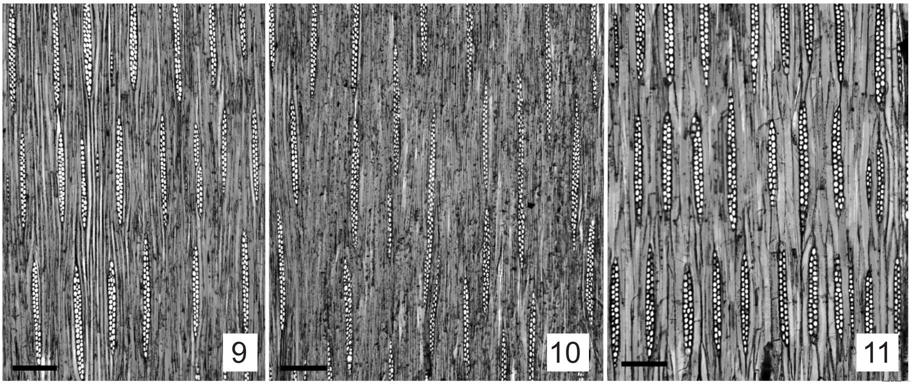

Figures 9–11. Longitudinal sections of *Caesalpinia echinata* wood sticks showing variation of storied rays. – 9: Storied rays in A-class sticks. – 10: Non-storied rays in C-class sticks. – 11: Storied rays in D-class sticks. — Scale bars: 100 µm.

Table 2. Mean values and standard deviation (SD) of wood anatomical features of the four classes of *Caesalpinia echinata* sticks. Distinct letters indicate significant differences at the $P < 0.05$ level (S-N-K or Dunn tests) among classes of sticks.

Wood anatomy features	Sticks			
	Class A	Class B	Class C	Class D
Vessels				
Frequency (mm^{-2})	13.1 ± 4.9 b	14.2 ± 4.4 b	16.7 ± 5.1 a	18.0 ± 5.6 a
Element length (µm)	353.7 ± 44.8 b	353.4 ± 49.0 b	341.4 ± 54.0 b	372.4 ± 52.2 a
Diameter (µm)	108.5 ± 20.0 a	105.4 ± 16.3 a	108.3 ± 17.9 a	108.5 ± 17.2 a
Fibers				
Length (µm)	1158.7 ± 242.1 a	1158.8 ± 243.3 a	1092.5 ± 233.6 b, c	1107.6 ± 219.6a,c
Diameter (µm)	17.3 ± 2.3 a	16.7 ± 2.1 b	16.9 ± 2.2 a,b	16.6 ± 2.1 b,a
Lumen diameter (µm)	5.1 ± 2.3 a	4.5 ± 2.2 a	4.4 ± 2.3 b	4.2 ± 1.9 b
Wall thickness (µm)	6.1 ± 1.3 a	6.1 ± 1.2 a	6.3 ± 1.1 a	6.2 ±1.0 a
Rays				
Height (µm)	229.8 ± 49.4 a	250.9 ± 64.6 b	230.3 ± 65.8 a	250.8 ± 57.2 b
Height (cells)	17.2 ± 3.8 b, a	18.9 ± 5.1 a	16.9 ± 4.8 b	17.8 ± 3.9 a
Width (µm)	18.7 ± 4.6 b	18.7 ± 4.2 b	20.3 ± 5.4 a	20.0 ± 4.8 a,b
Width (cells)	2.2 ± 0.5 a	2.3 ± 0.5 a	2.3 ± 0.6 a	2.2 ± 0.4 a
Frequency (mm)	10.2 ± 1.5 b	10.6 ± 1.6 b	11.3 ± 1.6 a	11.6 ± 1.2 a

Table 3. Cellular components percentages of Pernambuco wood of class A and D.

Cellular components	Class A	Class D
Parenchyma (%)	18.1	28.3
Vessels (%)	16.9	24.5
Fibers (%)	58.8	40.9
Rays (%)	6.0	6.1

Table 4. Physical, mechanical and acoustic properties of Pernambuco wood sticks obtained by destructive tests. Data obtained following the Brazilian norms NBR 7190/97.

Class and stick number	Specific gravity (Kg m^{-3})	Modulus of elasticity (Kgf cm^{-2})	Modulus of rupture (Kgf cm^{-2})	Speed of sound (m s^{-1})
Class A – 4	1004	189,864	2114.5	4348
Class A – 5	1005	192,156	2044.5	4372
Class A – 6	1019	235,326	2449.2	4805
Class D – 1	973	168,178	1536.4	4157
Class D – 2	873	100,296	892.8	3389
Class D – 3	850	166,949	1723.9	4431

Comparing the proportion of cell elements among the A and D classes, the former had a lower percentage of vessels and parenchyma and higher percentage of fibers (Table 3).

Physical, mechanical and acoustic properties

A-class sticks had higher values of wood density, speed of sound propagation, modulus of elasticity and modulus of rupture. Wood density and the speed of sound propagation obtained by destructive and nondestructive methods (Table 1 and 4), were proportional. This indicates that these parameters would be useful in the selection of sticks for bow making.

Chemical properties

Content of lignin and extractives

The sticks from the A-class showed a higher percentage of Klason lignin and lower quantities of the hydrosoluble and ethanol/benzene soluble extractives (Table 5).

Table 5. Contents of Klason lignin and extractives of Pernambuco wood.

Class	Contents of lignin (%)	Hydrosoluble extractives (%)	Extractives soluble in ethanol/benzene (%)
A	70.3	22.6	22.3
D	60.4	25.0	26.3

Table 6. Pearson and Kendall correlation with ordination axes.

Features	Principal components Axes 1	Axes 2
Vessel frequency	0.289	-0.926
Vessel element length	0.693	-0.586
Ray frequency	0.458	-0.587
Ray height	0.169	-0.777
Ray width	0.105	-0.855
% Fibers	-0.835	0.498
Contents of lignin	-0.965	-0.178
Contents of extractives	0.835	-0.498
Density - 12%	-0.766	0.321
Velocity of propagation of sound	-0.930	-0.347
Modulus of elasticity	-0.976	-0.197
Modulus of rupture	-0.991	-0.114
Percentage of explained variation	75.42%	19.88%

Principal component analysis (PCA)

Table 6 and Figure 11 show the correlations among the variables examined and the first and the second ordination axes, responsible for 95.3% of the explanation of PCA. Axis 1 contributed 75.5% of the variability and the characteristics which were most correlated to it were vessel element length (r = 0.693), fiber percentage (r = -0.835), lignin content (r = -0.965), extractive content (r = -0.835), wood density at 12% (r = -0.766),

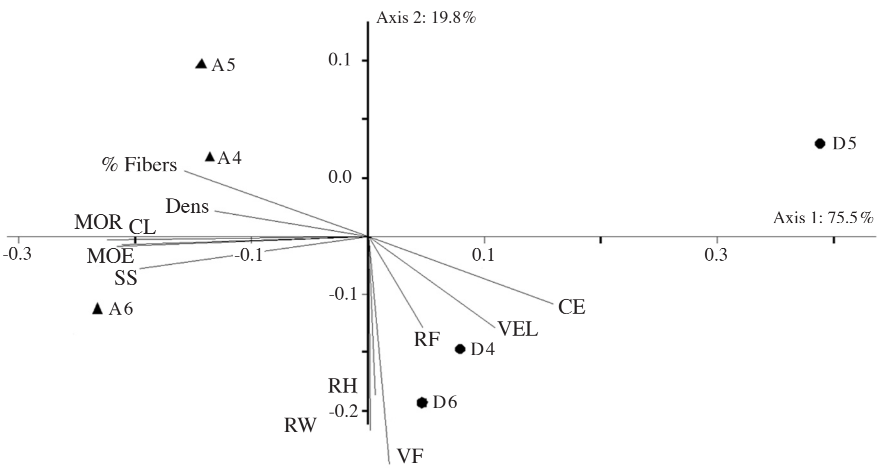

Figure 12. Ordination of A-class (▲) and D-class (●) sticks according to their anatomical, physical, mechanical, chemical and acoustical properties. Vessel frequency (VF), vessel element length (VEL), ray frequency (RF), ray height (RH), ray width (RW), percentage of fibers (% Fibers), contents of extractives (CE), contents of lignin (CL), density (Dens), speed of sound (SS), modulus of elasticity (MOE) and modulus of rupture (MOR).

speed of sound propagation (r = -0.930), modulus of elasticity (r = -0.976) and modulus of rupture (r = -0.991).

Axis 2 contributed with 19.8% of the variability, and the characteristics with higher correlation coefficients were: vessel frequency (r = -0.926), ray height (r = -0.777) and ray width (r = -0.855).

DISCUSSION

According to Wegst (2006), wood density is a very important property which influences other wood properties. The density changes in relation to the proportion of cells in the wood, their type and dimension, cell wall thickness and extractive content. Panshin and DeZeeuw (1964) and Rao *et al*. (1997) emphasize the importance of the proportion of different cell types in determining wood density and mechanical resistance. Other authors have confirmed the relationship between wood density and cell composition (Fujiwara *et al*. 1991; Fujiwara 1992; Green *et al*. 1999).

According to Lombardi (personal communication) the wood needs to have high density in order to produce bows with high quality. With high density sticks it is possible to obtain bows with the ideal weight and stiffness. Balanced, fine and light bows please the musicians because they can be handled more easily.

We observed that sticks with density ≥ 1000 kg m^{-3} at 12% humidity are ideal for the manufacture of higher quality bows. Nevertheless, sticks with much higher density than this value should be avoided because they will result in bows that are excessively thin, less stable and more susceptible to breakage.

The A-class sticks had a lower frequency of vessels and rays when compared to the other classes. The PCA analyses show that the vector which represents density is inversely correlated to the vectors that represent ray and vessel frequency, suggesting that these characteristics have an effect on wood density.

Modulus of elasticity (MOE) is another wood property that has to be considered in the selection of the sticks for bow manufacturing (Lucchi 1986; Follmann 1995). The MOE influences energy propagation and dissipation throughout the stick. A stick with high MOE loses less vibrational energy when the bow hair is pressed against the instrument strings. This allows the best use of the energy imposed by the musician, guaranteeing less physical effort to obtain the same result.

A-class sticks presented higher MOE (206,000 Kgf cm^{-2}, average of three sticks) when compared to those of the D class (145,000 Kgf cm^{-2}), which reinforces the importance of this characteristic in the determination of the potential quality of the stick. Unfortunately, a destructive method is necessary for its measurement, which makes it impossible to use the stick afterwards. This problem can be solved by indirectly calculating modulus of elasticity by using the relation MOE = $(\text{velocity})^2 \times$ density (Lucchi 1986).

Wood elasticity is strongly influenced by the fiber percentage (Basson 1987). The results obtained from the pernambuco samples confirmed this, since the sticks classified as A, when compared to the other classes, had the higher fiber percentage and also higher stiffness.

However, a high wood density value and MOE alone are not guarantees of bow quality. According to Askenfelt (2002), the way the wood density and the stiffness are distributed throughout the stick is important also. If these parameters are not uniform throughout the stick the balance and the bow playability will be compromised.

The laws of acoustics indicate that the capacity for sound propagation in the wood is linked to its stiffness, which in turn is affected by its density (Wegst 2006). Some authors have related the speed of sound propagation to the composition of the cell wall and to the cell types that compose the wood, especially fibers and vessels (Obataya *et al.* 2000; Bucur *et al.* 2002).

Resistance to rupture is a characteristic that has to be taken in consideration in the selection of wood for bows. Although high resistance is not necessarily required during the bow manufacturing process or during its first months of use, it is responsible for the higher or lower durability of the stick, which can rupture accidentally if it is dropped or as a result of internal cracks originating as a result of repeated tensioning. According to Follmann (1995) the bow head is very susceptible to breakage, since it is its most vulnerable region.

Comparing the mean values of modulus of rupture of the different pernambuco sticks, it can be seen that these were 2202.7 Kgf cm^{-2} (mean of three sticks) for A-class sticks and 1384.4 Kgf cm^{-2} for D-class sticks. The higher proportion of fibers provides higher bow stiffness and resistance to tension during stringing. Figure 11 shows that the referent vectors related to the fiber percentage and the modulus of rupture are quite close, which is a good indicator of their correlation.

Resistance to rupture also suffers from the influence of the position of the annual growth rings and of the rays, both involved in the propagation of cracks (Reiterer *et al.* 2002). Follmann (1995) believes that, in order to get more resistance to rupture in the process of bow manufacturing, the position of the annual rings in the blanks must be considered. Ideally the sticks should be cut radially. In the present study the influence of the position of the annual rings could not be assessed because the positions varied among the different classes of stick.

For Panshin and DeZeeuw (1964), in addition to the anatomical characteristics, the proportion of lignin and cellulose present in the cell walls is directly related to the wood tensile properties. Impregnation by lignin, a rigid, hydrophobic and degradation resistant material, provides protection and diminishes the possibility of cell damage. According to Bergander and Salmén (2002), the arrangement of the various wood polymers can influence the stiffness properties of the cell wall. Obataya *et al.* (1998) mention that the quantity of these polymers affects the dynamic properties of the wood.

Both the lignin content and the elasticity modulus values were the highest in the A-class sticks. In PCA analysis the proximity of the vectors which represent such characteristics shows a direct correlation between these two variables, corroborating the importance of this polymer in the wood stiffness.

According to Minato *et al.* (1997), Sakai *et al.* (1999), and Matsunaga *et al.* (2000), the extractive contents can influence the quality of the woods used in the bow manufacturing. These authors observed that the impregnation of extractives collected from pernambuco wood, especially the protosappanin B, diminished the loss tangent (tan δ) in other wood species. Tan δ is defined as the measurement of vibrational decay and according to Matsunaga *et al*. (1996) a stick with high elasticity modulus and low tan δ loses less vibrational energy when the hair is pressed against the instrument strings. These authors suggested the higher color intensity in the pernambuco wood to be related to the higher extractive contents and to the lower tan δ values. In practice, we did not observe any correlation between the pernambuco wood color and the quality of the bow stick, as the bow maker (Mr. Daniel Lombardi) produced bows of excellent quality from blanks made from regions close to the sapwood, thus presenting lighter colors. Furthermore, higher extractive contents were observed in D-class sticks, which were of proven low quality.

In conclusion, for the selection of pernambuco wood samples used for bow manufacture the density associated to the speed of sound propagation through the wood should be considered. Destructive tests, which made the use of the sticks to make bows impossible, showed that the sticks with density over 1000 kg m^{-3} and speed of sound propagation over 4300 m s^{-1} are of greater potential value in bow manufacturing. Nondestructive analyses revealed that values over 5300 m s^{-1} associated to density over 1000 kg m^{-3} indicate high potential.

Among the anatomical characteristics of the pernambuco wood, those linked to the vessels and fibers showed higher correlation with the bow quality, since they are related to the wood density and stiffness. Potentially better sticks had comparatively lower frequency of vessels and higher percentage of fibers.

In relation to other properties of the wood, we concluded that the elastic modulus (over 180,000 Kgf cm^{-2}) and modulus of rupture (over 2000 Kgf cm^{-2}) are indicative of the stick potential quality for bow making, since these two properties are related to stiffness and resistance to rupture. Sticks with high stiffness transmit the energy applied by the musician when playing the instrument more efficiently. In addition, this is important in relation to how much deformation the stick can take under tension without breaking.

Sticks with higher quality potential had a higher lignin content, which was related to wood stiffness. We did not find a relationship between the extractive quantity and the potential quality of the wood for bow making.

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